

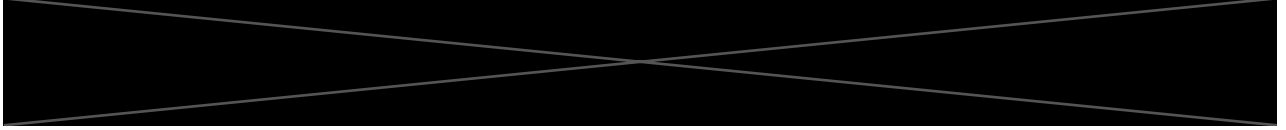
Neural Computation Assignment

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Neural Computation

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I Background - What we know & what we will be using

Before getting to the questions, here is a description for what we are working with. Stimulus reconstruction involves recreating or predicting sensory input that an organism experienced based on measured neural responses and spikes. The goal of this assignment is to recover the input stimulus from the spike train, r , of a leaky integrate and fire neuron.

- The stimulus is given by s . It is a T -dimensional vector of a zero-mean, Gaussian, lowpass filtered noise signal, with a standard deviation of `stimsd` and a correlation time of `stimtau`.
- The spike train r is a T -dimensional vector with a mean firing rate of:

$$\bar{r} = \frac{1}{T} \sum_{i=1}^T r_i \quad (1)$$

- The neuron also receives a background current of I_{mean} which sets its baseline firing rate.

- The reconstructed stimulus is:

$$\hat{s} = Rk + k_0 \quad (2)$$

where:

- k is a K -dimensional kernel vector to be determined.
- k_0 is a scalar offset.
- R is an $L \times K$ Matrix with entries $R_{ij} = r_{i-j+t_0} - \bar{r}$, where each row is a mean-subtracted snapshot of the spike train around one time point.
- $t_0 = \frac{K+1}{2}$ for odd K , centers the kernel window symmetrically, giving an acausal kernel.
- The reconstruction error is given by:

$$E = (s - \hat{s})^T (s - \hat{s}) \quad (3)$$

The task is to find the values of k and k_0 that minimise E .

II Question 1

Question:

Show that the reconstruction error, $E = (s - \hat{s})^2$, is minimised for a kernel $k = (R^T R)^{-1} R^T s$ and $k_0 = \bar{s}$.

Solution:

We want to show that minimising E gives the solutions $k = (R^T R)^{-1} R^T s$, and $k_0 = \bar{s}$.

We do that by setting $\frac{\partial E}{\partial k} = 0$ and $\frac{\partial E}{\partial k_0} = 0$ and computing the derivatives.

The reason why this works is because E is a quadratic function, so the point where the derivative is 0 is guaranteed to be the minimum, and not a saddle-point.

We can rewrite $E = (s - \hat{s})^2$ by substituting $\hat{s}$ for $Rk + k_0$. Now we get:

$$E = (s - Rk - k_0)^2 \quad (4)$$

or

$$E = \sum_i (s_i - [Rk]_i - k_0)^2 \quad (5)$$

We can solve for $\frac{\partial E}{\partial k_0} = 0$ first. Since we are taking the partial derivative with respect to k_0 , $s_i - [Rk]_i$ is a constant. Hence for simplicity, we can write the equation as:

$$E = \sum_i (c - k_0)^2 \quad (6)$$

where $c = s_i - [Rk]_i$.

Now we can differentiate the above using the chain rule. Now we get:

$$\frac{\partial E}{\partial k_0} = 2(c - k_0) \cdot (-1) \quad (7)$$

Substituting c for $s_i - [Rk]_i$, we get:

$$\Rightarrow \frac{\partial E}{\partial k_0} (s_i - [Rk]_i - k_0) = 2(s_i - [Rk]_i - k_0) \cdot (-1) \quad (8)$$

Setting $\frac{\partial E}{\partial k_0} = 0$, we get:

$$-2 \sum_i (s_i - [Rk]_i - k_0) = 0 \quad (9)$$

Dividing both sides by -2, we get:

$$\sum_i (s_i - [Rk]_i - k_0) = 0 \quad (10)$$

Now we can separate the sums and get:

$$\sum_i (s_i) - \sum_i ([Rk]_i) - \sum_i (k_0) = 0 \quad (11)$$

If we divide each term by the number of time points, we get their means:

$$\begin{aligned} \bar{s} - \bar{Rk} - k_0 &= 0 \\ k_0 &= \bar{s} - \bar{Rk} \end{aligned} \quad (12)$$

Since the entries of R is given by $R_{ij} = r_{i-j+t_0} - \bar{r}$, the $-\bar{r}$ part means we have subtracted the mean firing rate from every single entry of R . Because of this, each column of R has a zero-mean, therefore any weighted combination of them also has zero mean, regardless of what the weights k are. So $\bar{R}k = 0$ for any k . From this we finally get:

$$\boxed{k_0 = \bar{s}} \quad (13)$$

Now we can solve for $\frac{\partial E}{\partial k} = 0$. Firstly, we can write $E = (s - \hat{s})^2$ in the vector notation: $E = (s - \hat{s})^T (s - \hat{s})$. Note that since $k_0 = \bar{s}$, and the stimulus is zero-mean ($\bar{s} \approx 0$), k_0 can be dropped, and the equation for E can be simplified to:

$$E = (s - Rk)^T (s - Rk) \quad (14)$$

Now we can apply the transformation $(a - b)^T (a - b) = a^T a - 2a^T b + b^T b$ to the above equation, where $a = s$ and $b = Rk$. Following these rules:

$$a^T a = s^T s$$

$$2a^T b = 2s^T Rk$$

$$b^T b = k^T R^T Rk$$

This follows the transpose rule $(AB)^T = B^T A^T$

Now we get:

$$\Rightarrow E = s^T s - 2s^T Rk + k^T R^T Rk \quad (15)$$

Let's now take the partial derivative with respect to k for each term from the above equation.

$$\frac{\partial E}{\partial k} (s^T s) = 0$$

Treated as a constant

$$\frac{\partial E}{\partial k} (2s^T Rk) = 2R^T s$$

Break down of differentiation:

$s^T Rk$ is a scalar.

We can rewrite it by grouping $s^T R$ together.

This gives us $(R^T s)^T k$

This has the same form as $a^T k$ where $a = R^T s$

$a^T k$ is a dot product between a fixed vector a and k .

This follows the rule $\frac{\partial}{\partial b} a^T b = a$ for any fixed vector a .

$$\frac{\partial E}{\partial k}(k^T R^T R k) = 2R^T R k$$

Break down of differentiation:

$k^T R^T R k$ has the form $k^T a k$, where $a = R^T R$.

Since $R^T R$ is symmetric, we can apply $\frac{\partial}{\partial k}(k^T a k) = 2ak$.

Applying this to the term, we get $2R^T R k$.

So now we have the equation:

$$\frac{\partial E}{\partial k} = -2R^T s + 2R^T R k \quad (16)$$

We can set this to 0 and solve for k .

$$\begin{aligned} -2R^T s + 2R^T R k &= 0 \\ R^T R k &= R^T s \end{aligned} \quad (17)$$

We can solve for k by multiplying each side by $(R^T R)^{-1}$:

$$(R^T R)^{-1}(R^T R)k = (R^T R)^{-1}R^T s \quad R^T R)^{-1}(R^T R) = I \text{ (identity matrix)}$$

$$\boxed{\therefore k = (R^T R)^{-1}R^T s} \quad (18)$$

III Question 2

Question:

In particular, when firing rates are low, the neuron's response can be approximated by a Poisson process. Show that for a Poisson spike train $\langle (r(t_i) - \langle r \rangle)(r(t_j) - \langle r \rangle) \rangle = \rho \delta_{ij}$.

Breaking down the question: *What we know:*

- A Poisson process is a stochastic process in which discrete events occur randomly, independently, and at a constant average rate ρ .
 - In the context of a spike train, each time bin either contains a spike or not. if a time bin contains a spike $r(t_i) = 1$, and if not then $r(t_i) = 0$
 - Spikes are independent of each other. Whether a spike occurred at time t_i tells us nothing about whether one occurred at t_j .
 - the variance of the spike count equals the mean (key property of a Poisson process)

known as the Fano factor equaling 1).

- The equation $\langle (r(t_i) - \langle r \rangle)(r(t_j) - \langle r \rangle) \rangle = \rho \delta_{ij}$ is the autocovariance of the spike train.
 - $r(t_i)$ is the spike train value at time t_i - 1 if spike occurred, 0 if no spike occurred.
 - $\langle r \rangle$ is the expected mean firing rate - the same as $\bar{r}$.
 - $r(t_i) - \langle r \rangle$ is the mean-subtracted spike train at t_i , centering the spike train around 0.
 - ρ is the mean firing rate of the poisson process.
 - δ_{ij} is the Kronecker delta - equal to 1 when $i = j$ and equal to 0 when $i \neq j$.

Solution:

We know that $r(t_i)$ can either be 1 or 0. It is a Bernoulli random variable. The probability of a Bernoulli variable being 1 is given by:

$$\langle r(t_i) \rangle = p$$

$$\langle r(t_i)^2 \rangle = p \quad \text{since } 1^2 = 1; 0^2 = 0$$

The probability p of a spike occurring in a bin of size Δt is equal to $\rho \Delta t$. For small values of Δt , $\rho \Delta t \ll 1$.

Now we must consider 2 cases: when $i = j$ and when $i \neq j$.

Case 1: $i \neq j$:

We need to show that $\langle (r(t_i) - \langle r \rangle)(r(t_j) - \langle r \rangle) \rangle = \rho \delta_{ij}$. Let us look at the left and right hand sides of the equation separately.

Left Hand Side:

Multiplying the terms of the L.H.S, we get:

$$r(t_i)r(t_j) - \langle r \rangle r(t_i) - \langle r \rangle r(t_j) + \langle r \rangle^2$$

Now we can distribute the expectation across each term:

$$\langle r(t_i)r(t_j) \rangle - \langle r \rangle \langle r(t_i) \rangle - \langle r \rangle \langle r(t_j) \rangle + \langle r \rangle^2 \quad (19)$$

Since $\langle r(t_i) \rangle = \langle r(t_j) \rangle = \langle r \rangle$:

$$\begin{aligned} \langle r(t_i)r(t_j) \rangle - \langle r \rangle^2 - \langle r \rangle^2 + \langle r \rangle^2 \\ \Rightarrow \langle r(t_i)r(t_j) \rangle - \langle r \rangle^2 \end{aligned} \quad (20)$$

$r(t_i)$ and $r(t_j)$ are independent random variables. Hence we can use the independence rule.

$$\langle r(t_i)r(t_j) \rangle = \langle r(t_i) \rangle \cdot \langle r(t_j) \rangle \quad (21)$$

Since the Poisson process has a constant rate, the expected value is the same at every point.

$$\begin{aligned} \langle r(t_i) \rangle &= \langle r \rangle \\ \langle r(t_j) \rangle &= \langle r \rangle \\ \therefore \langle r(t_i) \rangle \cdot \langle r(t_j) \rangle &= \langle r \rangle^2 \end{aligned} \quad (22)$$

Substituting $\langle r(t_i) \rangle \cdot \langle r(t_j) \rangle$ for $\langle r \rangle^2$, we get:

$$\langle r \rangle^2 - \langle r \rangle^2 = 0 \quad (23)$$

Now considering the right hand side $\rho\delta_{ij}$ for when $i \neq j$:

We already know that $\delta_{ij} = 0$ when $i \neq j$. Hence $\rho\Delta t = 0$. This proves the equation:

$$\boxed{\langle (r(t_i) - \langle r \rangle)(r(t_j) - \langle r \rangle) \rangle = \rho\delta_{ij}} \quad \text{for } i \neq j \quad (24)$$

Case 2: $i = j$:

When $i = j$, we are looking at the same time point. So we can use the term t_i for both time points and write $\langle r(t_i) \rangle \cdot \langle r(t_i) \rangle$ as $\langle r(t_i)^2 \rangle$. Now we get the equation:

$$\langle r(t_i)^2 \rangle - \langle r \rangle^2 \quad (25)$$

Earlier, we established that the probability of $\langle r(t_i)^2 \rangle$ and $\langle r(t_i) \rangle$ will be either 1 or 0. And $\langle r(t_i) \rangle$ can be expressed as $\langle r \rangle$. So we can rewrite the above equation as:

$$\langle r \rangle - \langle r \rangle^2 \quad (26)$$

Expressing this as probabilities, we get:

$$\begin{aligned} \rho\Delta t - (\rho\Delta t)^2 \\ \rho\Delta t(1 - \rho\Delta t) \end{aligned} \quad (27)$$

For small time bins, $\rho\Delta t \ll 1$.

$$\therefore (1 - \rho\Delta t) \approx 1.$$

$$\Rightarrow \rho\Delta t(1 - \rho\Delta t) = \rho\Delta t$$

In discrete time, the spike train $r(t_i)$ is equal to the number of spikes divided by Δt . $\therefore \Delta t$ cancels out, leaving just ρ .

On the right hand side, we have $\rho\delta_{ij}$. When $i = j$, $\delta_{ij} = 1$. Hence $\rho\delta_{ij} = \rho$. This proves the equation:

$$\boxed{\langle (r(t_i) - \langle r \rangle)(r(t_j) - \langle r \rangle) \rangle = \rho\delta_{ij}} \quad \text{for } i = j \quad (28)$$

IV Question 3

Question:

Implement the reconstruction algorithm using a long enough (acausal) window. Plot the kernel k both using the above expression and under the assumption that the response approaches a Poisson process. Explain the difference.

Solution:

The difference between the kernel formula and the Poisson approximation kernel is related to the temporal correlations. The reconstruction kernel $k = (R^T R)^{-1} R^T s$ makes no assumptions about the structure of the spike train. It accounts for the full temporal correlations of the neural response through the $(R^T R)^{-1}$ term. The Poisson approximation kernel assumes that the spike train has no temporal correlations, and that spikes are independent of each other.

We can derive the Poisson kernel directly from the result of Question 2. The (j, k) entry of $R^T R$ is a sum over T time points of products of mean-subtracted spike counts at lag $j-k$ apart. This equals T times the autocovariance at lag $j-k$. From Question 2, the autocovariance of a Poisson spike train equals $\rho\delta_{jk}$, so $R^T R \approx \rho T \cdot I$. Substituting this into the exact kernel formula, and using $\rho T = n_{spikes}$, gives:

$$k_{Poisson} = \frac{1}{n_{spikes}} R^T s \quad (29)$$

This derivation rests on the assumption that the autocovariance is zero at all non-zero lags (spikes are truly independent). Figure 1 shows the normalised autocorrelation of the spike train. The dip just after lag zero reflects the refractory period: immediately after a spike

fires, the membrane resets and another spike is suppressed for a few milliseconds. This means spikes are negatively correlated at short lags, so the off-diagonal entries of $R^T R$ near lag zero are negative rather than zero. Therefore the Poisson assumption is not satisfied here.

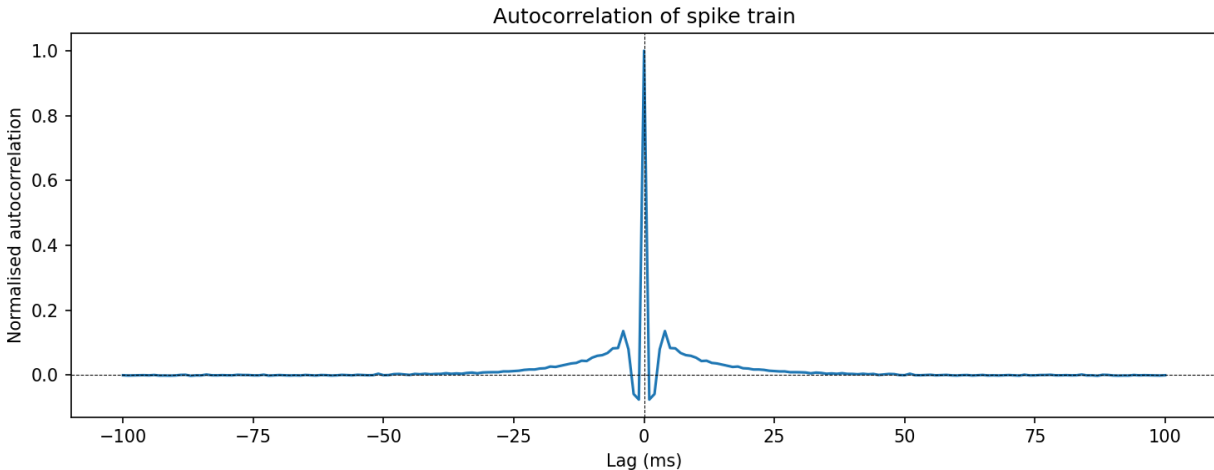


Figure 1: Caption

Figure 2 shows both kernels plotted against time lag. The exact kernel drops sharply nearly to 0 within the first 10ms after the peak. The Poisson kernel decays gradually and smoothly. This difference can be explained due to the refractory period as mentioned above. The exact kernel accounts for the negative off-diagonal structure of $R^T R$ through the full matrix inverse, which compensates by up-weighting very recent spikes more strongly. The Poisson kernel spreads weight more broadly across past lags and produces the wider, slower-decaying shape as seen in the figure. The Poisson approximation improves at lower firing rates, where spikes are rarer and less likely to fall close together in time, making the refractory period less consequential.

V Question 4

Question: Reconstruct the stimulus, and plot the original stimulus and the reconstruction. In addition, calculate the normalized reconstruction error $e = \sqrt{E}/std(s)$. Comment on the reconstruction quality.

Solution: As suggested in the hint, I replaced the neural response with a linear mapping ($r(t_i) = s(t_i)$) to verify the reconstruction algorithm before applying it to the spike train. In this case the stimulus and response are identical as shown in figure 3.

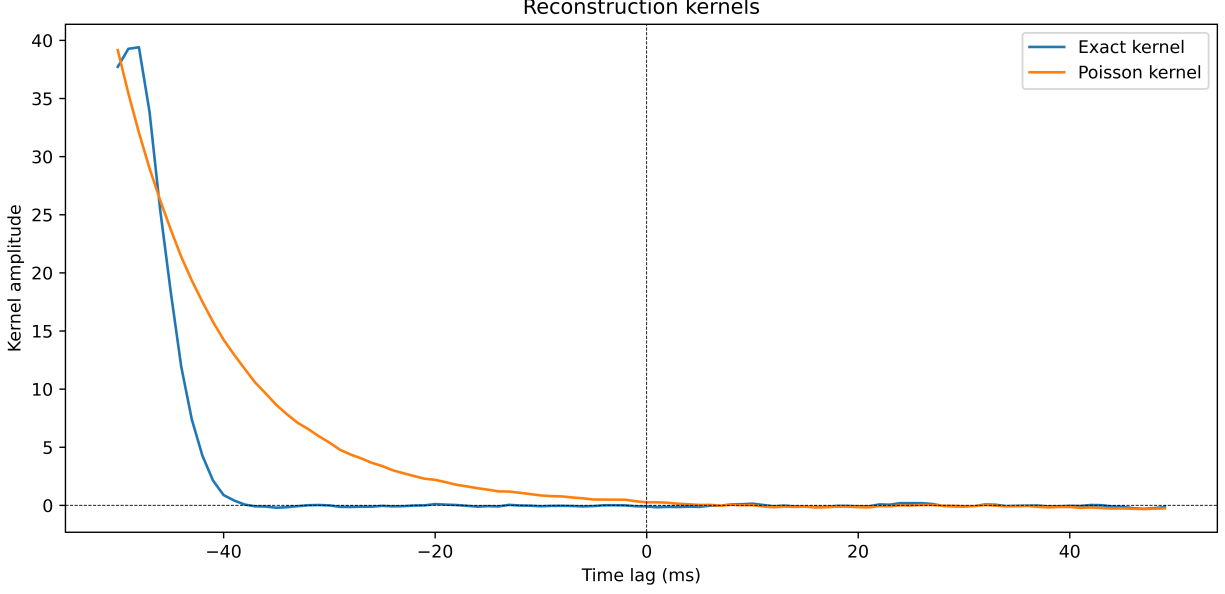


Figure 2: Exact and Poisson Reconstruction Kernels

Now we can move to the stimulus reconstruction using the exact kernel and Poisson kernel. Figure 4 shows the stimulus reconstruction using the exact and Poisson kernels.

Table 1 shows the stimulus parameters and neuron parameters to do the stimulus reconstruction:

The **Normalised reconstruction error** was calculated for both kernels using the formula $e = \sqrt{E}/std(s)$:

Normalised reconstruction error (exact): 0.7253

Normalised reconstruction error (Poisson): 0.8515

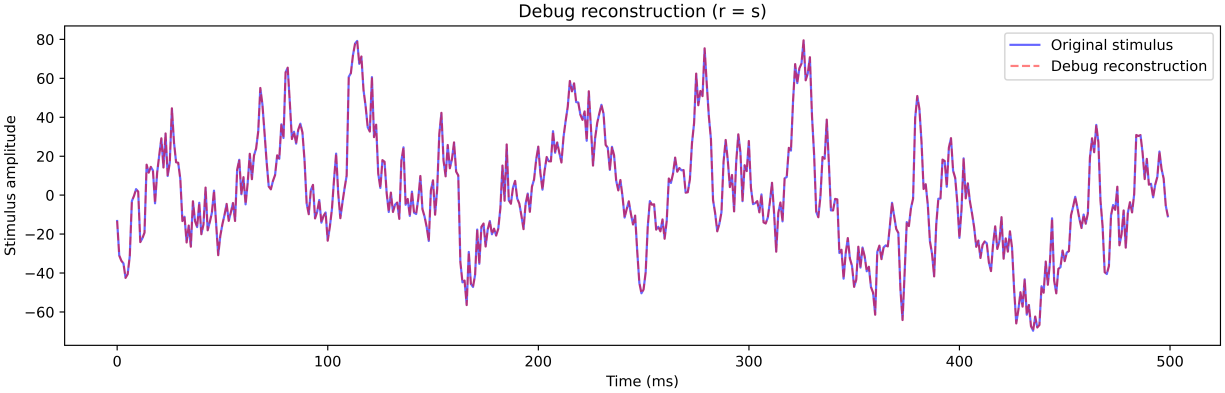


Figure 3: linear mapping to verify reconstruction algorithm

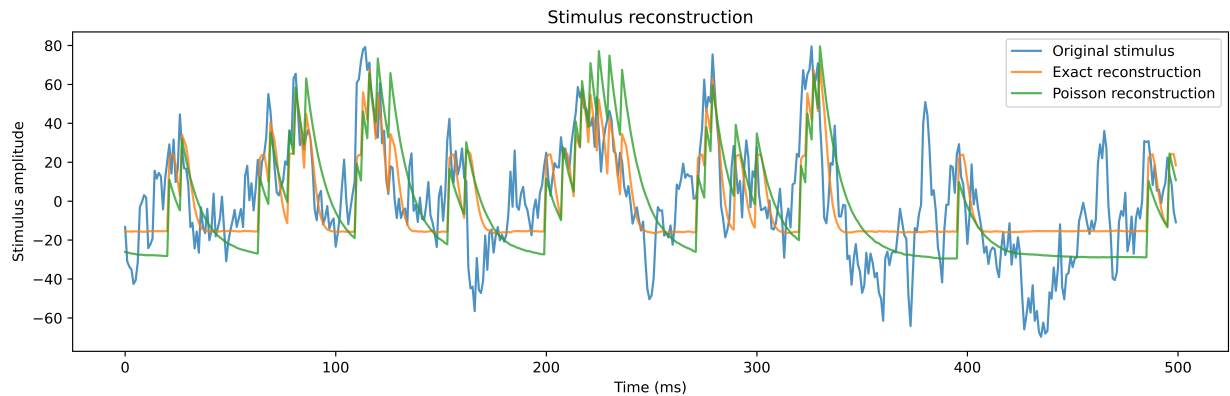


Figure 4: Stimulus reconstruction with exact and Poisson kernels

Parameter	Value
stim_sd	30
stim_tau	10
vrest	-70
vreset	-70
vthr	-50
tau_mem	10
imean	20

Table 1: Stimulus & neuron parameters for simulation and reconstruction

Reconstruction Quality:

Figure 4 shows the original stimulus (blue), the exact reconstruction (orange), and the Poisson reconstruction (green) over the first 500 ms of the simulation. Here is what we can discern from the figure:

- **Phase:** Both reconstructions are shifted slightly to the right relative to the original stimulus. The peaks and troughs of the reconstructed signals lag behind the true stimulus by approximately 10 – 20 ms. The lag is consistent with the membrane time constant $\tau_{mem} = 10$ ms.
- **Amplitude:** Both reconstruction kernels capture the broad peak amplitudes reasonably well, though with some differences in accuracy. The exact reconstruction (orange) tracks the peak amplitudes closely in several places, reaching near the true stimulus values around $t = 100$ ms, $t = 330$ ms, and $t = 370$ ms, though it occasionally misses the tip of fast peaks. The Poisson kernel overshoots above the true stimulus at some peaks and descends too steeply after them, reflecting its broader kernel integrating

over a longer history of spikes and overcorrecting in both directions. At the minima, both reconstructions fail to follow the stimulus downward, instead plateauing at approximately -20mV regardless of how negative the true stimulus becomes. This occurs because with $I_{mean} = 20$ mV the steady state membrane potential sits exactly at threshold, so strongly negative stimulus values push the membrane below threshold, the spike train carries no information about the stimulus during these periods. With no spikes to work with, both kernels return to the mean of the stimulus.

The reconstruction quality can be understood in terms of the neuron's operating point. With $I_{mean} = 20$ mV, the steady state membrane potential is exactly at threshold, so the neuron fires when the stimulus is positive but goes silent when negative, since they push the membrane below threshold. This creates an asymmetry.

The exact kernel achieved a normalised reconstruction error of $e = 0.7253$ (≈ 0.73), recovering approximately 47% of stimulus variance. The Poisson kernel performs worse at $e = 0.8515$ (≈ 0.85), recovering only 28% of variance.

Both values are below 1, which means that both reconstructions are performing better than predicting the stimulus mean at every time point. The exact kernel outperforms the Poisson kernel because it accounts for the full temporal structure of the spike train. The remaining error in both kernels reflects two fundamental limitations of the leaky integrate and fire neuron as an encoder. The membrane time constant $\tau_{mem} = 10$ acts as a low-pass filter. So fast fluctuations in the stimulus are smoothed out before they can drive the neuron to fire. Additionally the thresholding operation discards all information about subthreshold activity entirely. When the membrane does not reach the voltage threshold (v_{thr} , no spike is generated and the stimulus value at that moment leaves no trace in the spike train.

VI Question 5

Question:

Research how the background current influences the kernel shape and reconstruction error and interpret your findings.

Solution:

I_{mean} represents the constant background current that a neuron receives from the rest of the network, in the absence of a specific stimulus of interest. When I_{mean} is low, the neuron

receives little background current and is far below threshold at rest. When I_{mean} is high, the neuron receives strong background current which could be just below or even above the threshold at rest.

By doing a parameter sweep of I_{mean} and plotting the values against the reconstruction error, we can understand how the background current affects the reconstruction. Figure 5 shows the reconstruction error for the exact and Poisson kernels for I_{mean} values ranging from 10 to 140 mV. We can see that the exact kernel has the lowest reconstruction error at

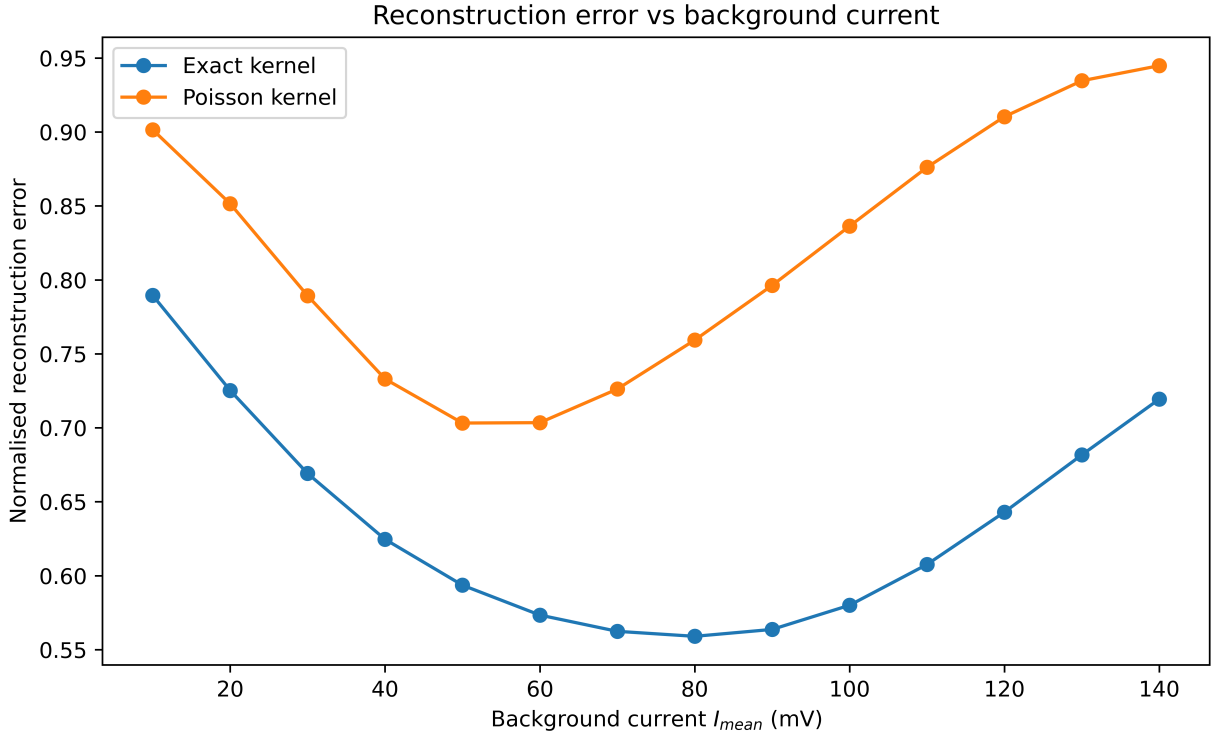


Figure 5: Normalised reconstruction error for the exact and Poisson kernels as a function of background current swept from 10 to 140 mV in steps of 10

$I_{mean} = 80$ and the Poisson kernel has the lowest reconstruction error when $I_{mean} \approx 50$ to 60. As the value of I_{mean} increases (till 80), the higher background current pushes the membrane closer to and above threshold. The neuron fires more frequently. More spikes means more information, so reconstruction improves. After the kernels reach their optimal point, the high background current causes the neuron fires so frequently that it approaches saturation. The neuron fires almost regardless of what the stimulus is, so individual spikes carry less information about the stimulus and more about the background current. The reconstruction kernel can no longer distinguish between stimulus-driven spikes and background spikes, and

reconstruction quality goes down.

Effect on reconstruction kernel shape

Figure 6 shows the kernel shape for exact and Poisson kernels for I_{mean} values ranging from 10 to 90 in steps of 20.

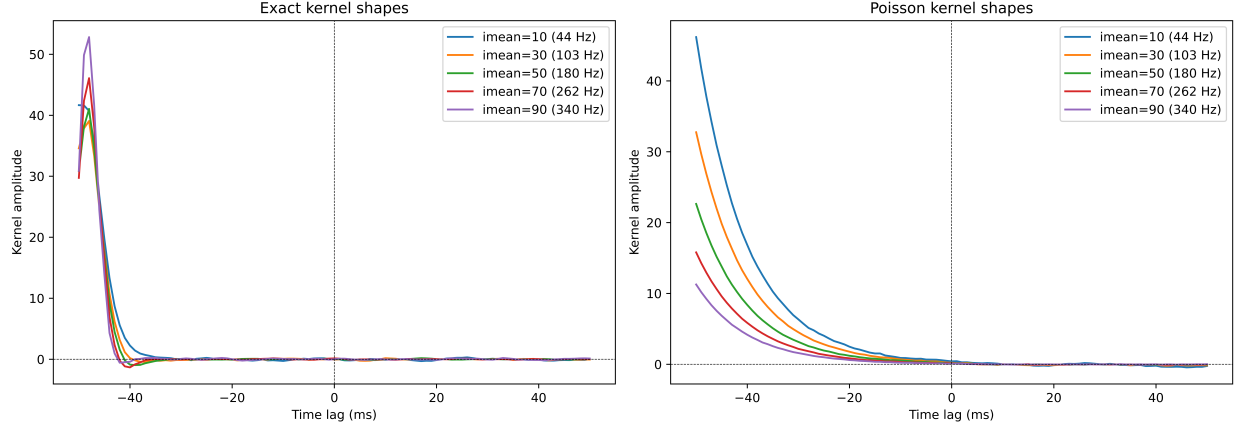


Figure 6: Kernel shapes of Exact kernel (left) and Poisson kernel (right) with varying I_{mean} values: $I_{mean} \in 10, 30, 50, 70, 90$

The **Exact kernels** are shown on the left side of figure 6. All kernels for the ranging values of I_{mean} share the same shape, a sharp peak at time lag ≈ -50 to -45 followed by a sharp drop to amplitude ≈ 0 . The peak location does not shift with I_{mean} , meaning the dominant integration window of the neuron stays fixed regardless of background current. This is because the peak is determined by $\tau_{mem} = 10$ ms and the kernel window K which are both fixed.

The peak amplitude shows a non-monotonic pattern. The higher amplitudes at $I_{mean} = 70$ and $I_{mean} = 90$ reflect the fact that each individual spike needs to be weighted more heavily in the reconstruction to account for the higher baseline firing rate. At high firing rates, many spikes occur even when the stimulus is near zero, so the kernel must assign a larger weight per spike to extract the stimulus-driven component from the background activity.

The small negative dip just after the peak is visible for all curves and is more pronounced at higher I_{mean} values. This reflects the refractory period: after a spike fires, another spike is suppressed, creating a brief negative correlation in the spike train. The exact kernel captures this through $(R^T R)^{-1}$ and compensates by slightly down weighting the contribution of spikes that follow very closely after a spike.

The **Poisson kernels** are shown on the right side of figure 6. They are all smooth exponentially decaying curves with no sharp peak or negative dip, consistent with the Poisson assumption that spikes are independent and the kernel is simply a scaled version of the spike-stimulus cross-correlation $\frac{1}{n_{spikes}} R^T s$. The peak amplitude decreases monotonically as I_{mean} increases. This is due to the $\frac{1}{n_{spikes}}$ normalisation. As I_{mean} increases, the neuron fires more frequently (n_{spikes} increases) and the entire kernel is scaled down proportionally. The shape of the decay remains the same for all I_{mean} values. The Poisson kernel assigns a smaller weight per spike as firing rate increases, which is in the correct direction but the wrong magnitude, since it does not account for the covariance structure that the exact kernel handles through $(R^T R)^{-1}$